

Project overview - Tampines precinct thermal load and cooling simulation

The Tampines precinct project aims to develop EnergyPlus models of selected HDB residential blocks of a precinct in Tampines to study their **indoor thermal conditions** and **air-conditioning cooling demand**.

The main work included constructing floorplans and representing each room as individual EnergyPlus thermal zones, followed by simulating the cooling required to maintain a predetermined cooling setpoint of 25°C at selected rooms. I compared cooling demand and indoor temperature under different:

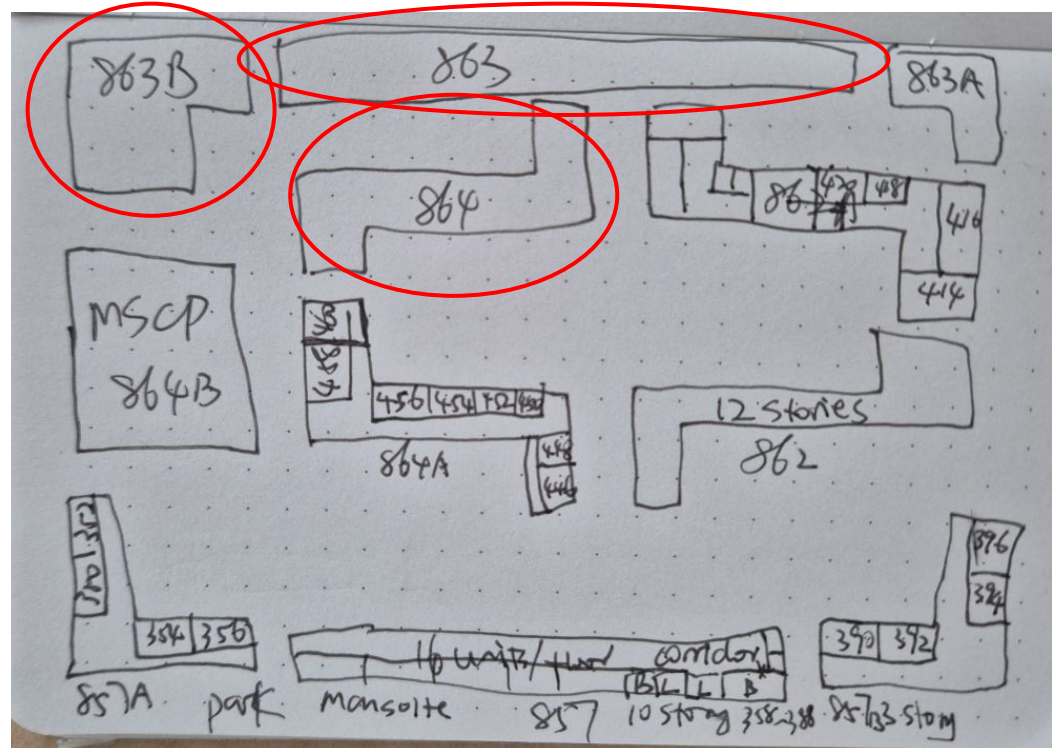
- Wall material configurations
- AC operating schedules
- Room, unit and storey levels

Developed Python scripts to process EnergyPlus outputs and generate comparison plots.

HDB Blocks modelled

For entire precinct:

- Blk 863B mirrored 4 times
- Blk 864 mirrored 4 times
- Blk 863 mirrored 2 times



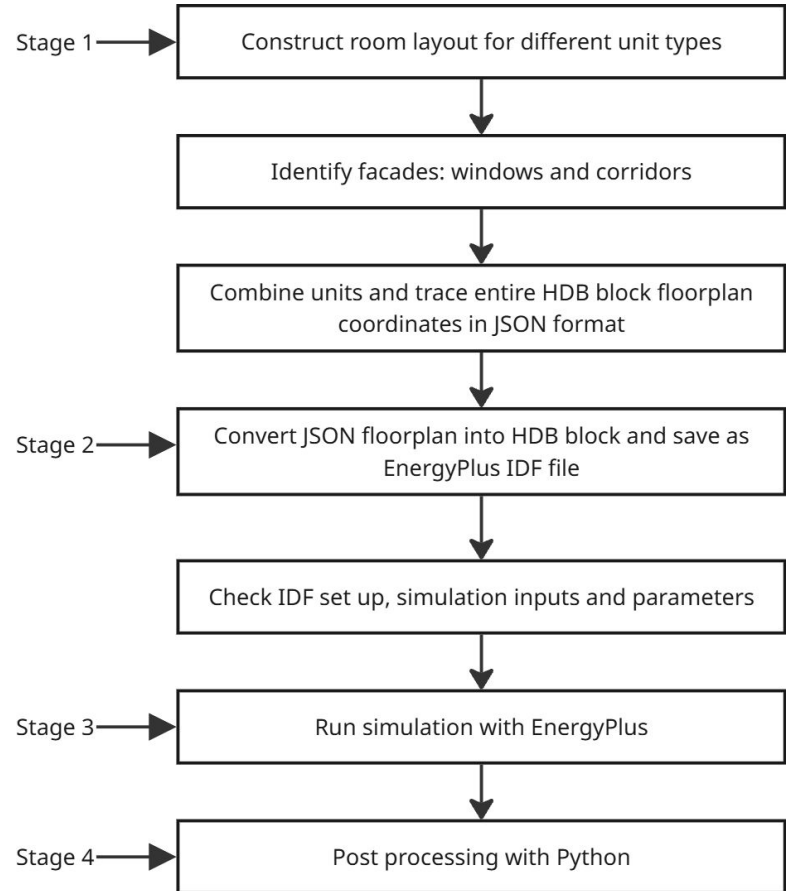
Approach

Input

- Unittemplate.JSON: Room layout of various unit types (4R, 5R, corner, corridor, point block unit etc)
- IDF file of constructed blocks with simulation parameters (wall material, thermostat setpoint, AC settings, infiltration, sun and wind assignment, etc)

Output

- CSV files of containing Total, Sensible and Latent cooling energy, Zone mean air temperature
- Graphs and diagrams of annual totals and hourly trends

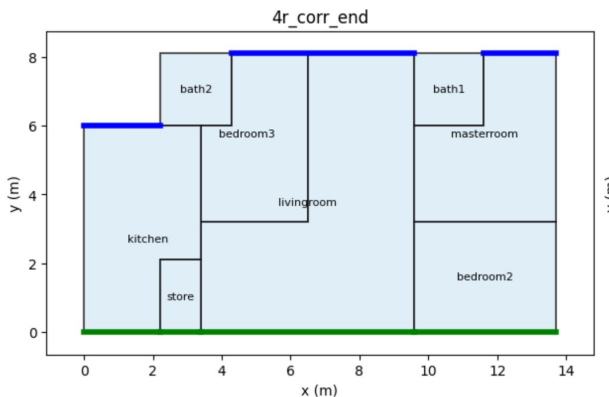


Stage 1: Create digital floorplan in JSON format

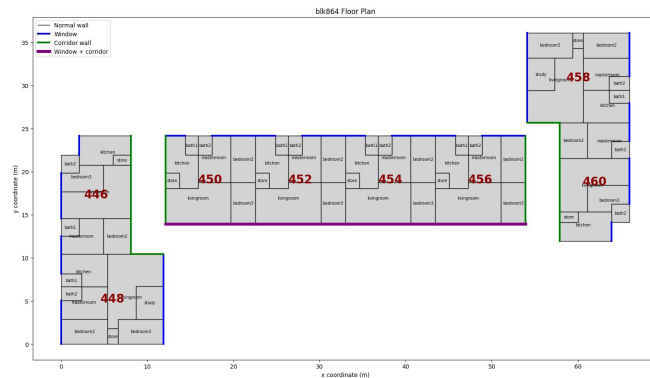
- Looked for room floor plan of different types of HDB units from various online sources (PropertyGuru, 99.co or HuiZhe)
 - different unit types: 3 room, 4 room, 5 room units
 - different unit types: corridor, corridor end, point block corner
 - units varied for different types of HDB (slab blocks, point blocks, ,executive maisonette etc)
- Traced room coordinates and converted into JSON coordinates for each unit
- Identified and recorded windows and corridors for each unit
- Created different mirrors of the unit (mirror about x, y and origin) as `unittemplate_mirrors.json`
- Coded a script that extracts required rooms, rotates/translate/combine in HDB block floorplan

```
"4r_corr_end": {  
  "rooms": {  
    "bath2": {"coordinates": [[2.2,8.1],[2.2,6.0],[4.3,6.0],[4.3,8.1]]},  
    "bedroom3": {"coordinates": [[4.3,8.1],[4.3,6.0],[3.4,6.0],[3.4,8.1]]},  
    "livingroom": {"coordinates": [[6.5,8.1],[6.5,3.2],[3.4,3.2],[3.4,8.1]]},  
    "bath1": {"coordinates": [[9.6,8.1],[9.6,6.0],[11.6,6.0],[11.6,8.1]]},  
    "masterroom": {"coordinates": [[11.6,8.1],[11.6,6.0],[9.6,6.0],[9.6,8.1]]},  
    "bedroom2": {"coordinates": [[9.6,3.2],[9.6,0.0],[13.7,0.0],[13.7,3.2]]},  
    "kitchen": {"coordinates": [[0.0,6.0],[0.0,0.0],[2.2,0.0],[2.2,6.0]]},  
    "store": {"coordinates": [[2.2,2.1],[2.2,0.0],[3.4,0.0],[3.4,2.1]]},  
  },  
  "facades": [{"start": [0.0,6.0], "end": [2.2,6.0], "type": "window", "wWR": 1}],  
}
```

1) JSON coordinates



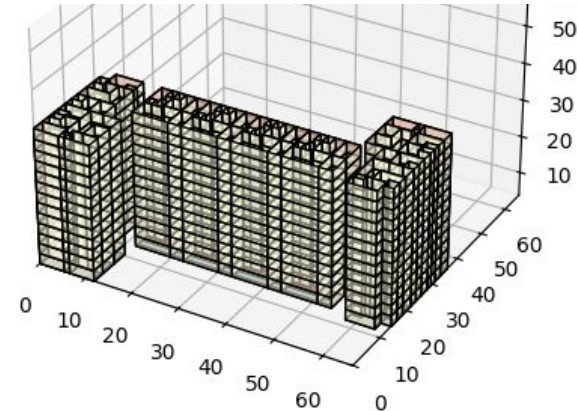
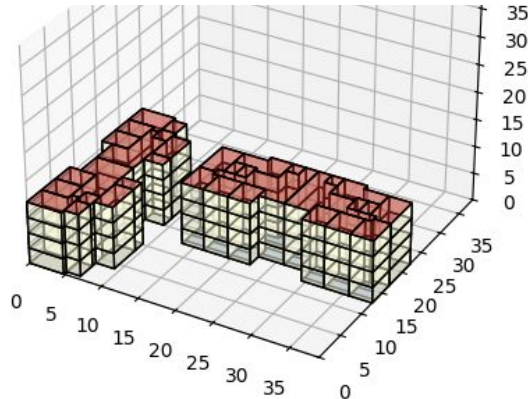
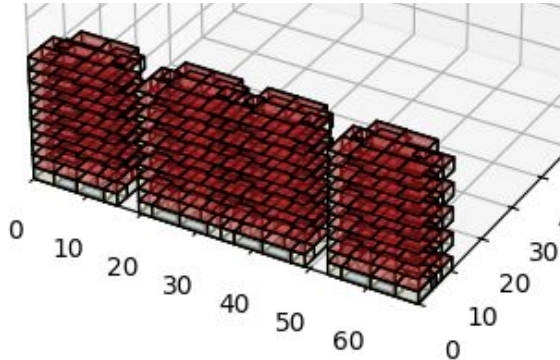
2) Unit floorplan



3) Block floorplan

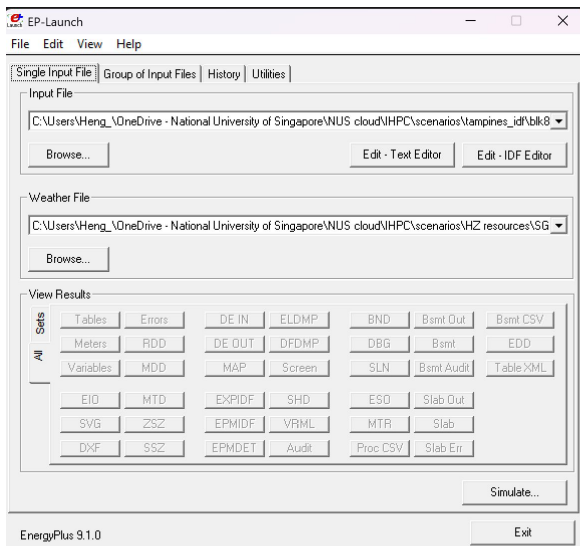
Stage 2: Convert JSON floorplan into EnergyPlus IDF

- Python script reads block floorplan JSON and constructs IDF file (void deck 3.6m, room height 2.8m)
- User must define simulation conditions and parameters:
 - Thermostat set point 25 deg, only cooling no heating
 - Max humidity 60%, only dehumidify no humidify
 - Aircon schedule: Always 2200 to 0700 (no special weekends and hols), then 24hrs
 - AC application where normal schedule is 2200hrs - 0700hrs. 24hr AC means AC on for 24 hrs
 - Infiltration: Bedrooms/master rooms: 0.2 ACH; Living 0.2 ACH when AC on or 1.0 ACH when AC off, others 1.0 ACH
 - Applies wall conditions based on window or facade (sun, noSun, wind, noWind)
- 5 material cases



Stage 3: Run Simulation

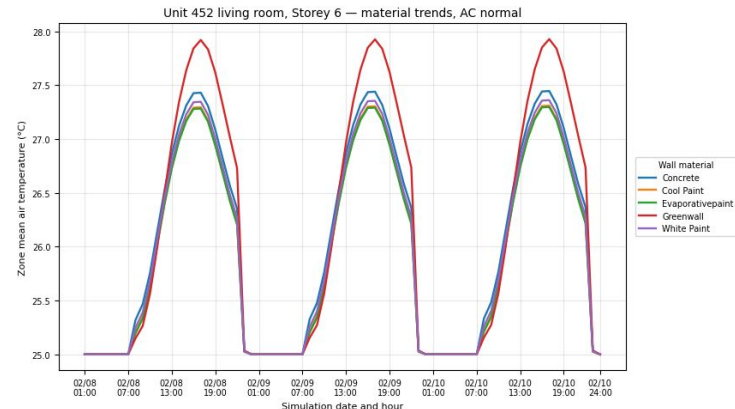
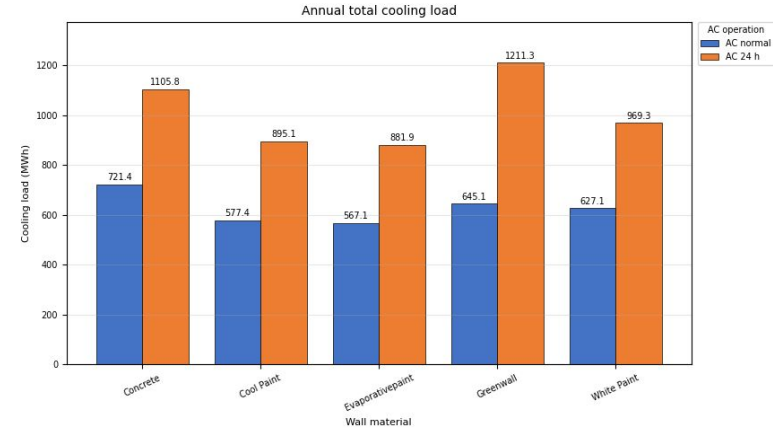
- Ran manually so I can check through results and errors to make sure everything is correct
 - but can be python scripted and automated with `subprocess.run()`
 - captures return code, eplusout.err, eplusout.csv
- Check error file for failures
- Ensure correct weather file is selected
- Each block has **20 runs** (2 AC schedules x 5 materials x 2 weather files) - up to 1000s for complex blacks (864)



```
command = [  
    energyplus_exe_path,  
    "-w",  
    epw_path,  
    "-d",  
    output_dir,  
    "-p",  
    output_prefix,  
    "-x",  
    "-r",  
    idf_path,  
]
```

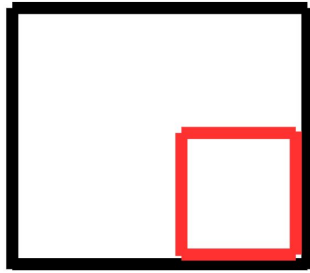
Stage 4: Post processing and analysis

- Compared cooling loads across:
 - 5 different wall materials
 - 2 different AC schedules (evening vs 24hr)
 - 2 different weather dates (3 days vs annual)
- Outputs:
 - Supply total cooling energy (annual and monthly)
 - Supply latent cooling energy (annual and monthly)
 - Supply sensible cooling energy (annual and monthly)
 - Living room mean zone temperature (hourly)
- Convert EnergyPlus CSV output into graphs and plots (as well as unit conversion from J to MWh)
- Compared annual cooling loads and hourly room temperature to study weather trends

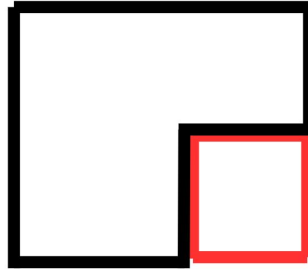


Key lessons (things I wish I knew earlier)

- `intersect_match()` issues – Room polygons must not overlap. This means when creating JSON outlines, no room should be inside or overlap another room



Overlapping - No



Separated - Yes

- Don't just blindly run simulations, take time to understand the idf file and its different sections so you know where and what to check before running simulation
- Larger blocks such as blk863_EM and blk864 takes some time to run. 400+ seconds and 1000+ seconds per simulation
- 1st floor (void deck) doesn't count for blk 863 and 864. If real HDB has 13 floors, only input 12 in JSON file. Only blk 863b has no void deck. *[Blk864 has 1 extra floor which led to about 7% more energy demand (I only found out about this quite late and didnt have time to rerun entire simulation)]*

Current Progress (handover to next intern)

- Constructed all block layout and ran individual block simulations
- Have not constructed the entire precinct yet, only completed all individual blocks. Next step would be to combine everything into 1 precinct

Future Work

- Create precinct – copy, mirror and translate the 3 key blocks (blk863, 863b and 864)
- Explore and compare different levels of fidelity (can higher or lower) to find optimal tradeoff between high and low fidelity
- Simulation automation

Assumptions:

- Assumed all corridors are blocked by sun (in reality, parapet wall doesn't fully block wall from sun)
- Removed balcony and extended it to living room (balcony itself is not a room because its outdoor exposed, but it also provides some shading effect to living room. I simplified by removing it and combining into living room)
- Executive Maisonette stairwell hole removed – both stories assumed to be 2 independent air zones instead of 1 large air circulating zone
- Assumed no internal loads (no people, no light and equipment)

Limitations:

- EvaporativePaint and GreenWall evaporative features not captured
- Constructed micro-blocks due to removal of corridors (hard to create connected corridors)